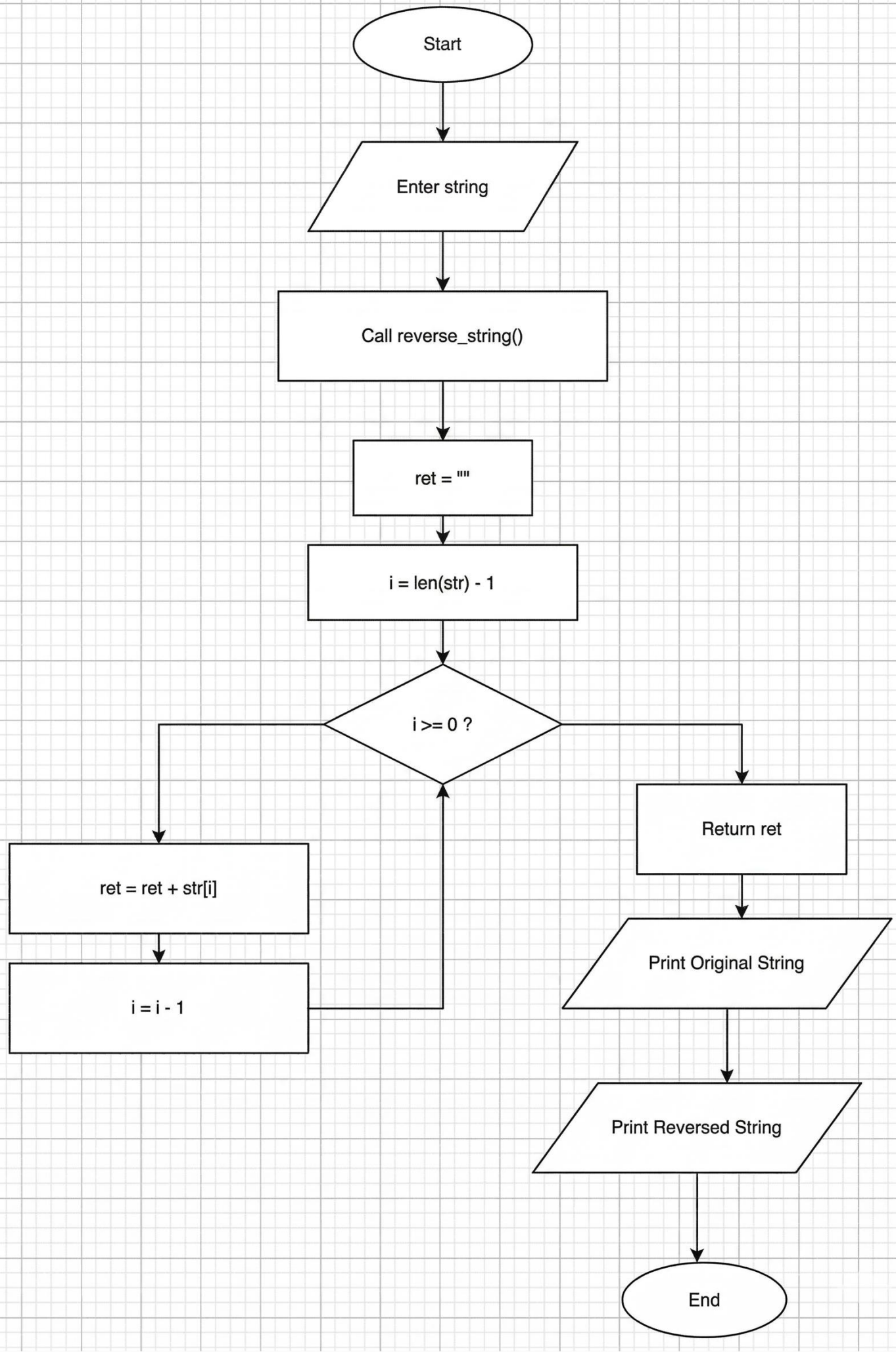
Experiment 10

10.1.2 Reverse String

Algorithm :

Step 1 : Start
Step 2 : Input string and store in user_string
Step 3 : Set result as an empty string
Step 4 : Find length of user_string and store in l
Step 5 : Set i = l - 1
Step 6 : Repeat steps while i is greater than or equal to 0
Step 7 : Get the character at index i from user_string
Step 8 : Add the character to result
Step 9 : Decrease i by 1
Step 10 : Go back to Step 6
Step 11 : Print result
Step 12 : Stop

Flowchart :


Start
Enter string
Call reverse_string()
ret = ""
i = len(str) - 1
i >= 0 ?
ret = ret + str[i]
i = i - 1
Return ret
Print Original String
Print Reversed String
End

Code :

```
def reverse_string(strl):
    res = ""
    ln = len(strl)
    for i in range(ln-1,-1,-1):
        res=res+strl[i]
    return res
user_input = input("Enter a string: ")
result = reverse_string(user_input)
print(f"Original String: {user_input}")
print(f"Reversed String: {result}")
```

Execution :

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10.1.2. Reverse a String

08:22

Write a Python program that prints the reverse of a given input string. The program uses a **reverse_string()** function that takes a string as input and returns its reversed string.

Input Format:

- The program prompts "Enter a string: " and reads a string representing the string to be reversed.

Output Format:

- The first line prints the original string, in the format:

```
Original String: <input_string>
```

- The next line prints the reversed string, in the format:

```
Reversed String: <reversed_string>
```

Sample Test Cases +

Explorer

ReverseS...

Submit

```python
def reverse_string(strl):
    res = ""
    ln = len(strl)
    for i in range(ln-1,-1,-1):
        res=res+strl[i]
    return res
user_input = input("Enter a string: ")
result = reverse_string(user_input)
print(f"Original String: {user_input}")
print(f"Reversed String: {result}")
```

Debugger

Average time	Maximum time
0.003 s	0.003 s
2.50 ms	3.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 3 ms Debug

Expected output	Actual output
Enter a string: PythonProgrammming	Enter a string: PythonProgrammming
Original String: PythonProgrammming	Original String: PythonProgrammming
Reversed String: gnimmmargorPnohtyP	Reversed String: gnimmmargorPnohtyP

Test case 2 3 ms

Terminal Test cases

Prev Reset Submit Next